

DATA ANALYTICS MAIN PROJECT

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Title : Netflix Analytics : Exploring Content Growth
and Patterns

Project Domain : Social Media & Public Sentiment

Dataset Year : Data collected from Up to 2025

Submission Date : 29-04-2026

Mentor Name : Kumaran M

Raw Dataset Link : <https://www.google.com/url?q=https%3A%2F%2Fwww.kaggle.com%2Fdatasets%2Fbhargavchirumamilla%2Fnetflix-movies-and-tv-shows-till-2025%2Fdata>

Cleaned Dataset : <https://docs.google.com/spreadsheets/d/1qnOb-P5hrZiOnCqt1jxQTCyL2BseBB2D/edit?usp=sharing&ouid=104138233421140273365&rtpof=true&sd=true>

NETFLIX ANALYTICS: EXPLORING CONTENT GROWTH AND PATTERNS

1. Project Title

Netflix Analytics : Exploring Content Growth and Patterns

2. Objective(Problem statment)

This project aims to analyze Netflix content data to identify meaningful patterns, trends, and insights using exploratory data analysis (EDA).

The objective is to understand how Netflix content has grown over time and to explore patterns in genres, ratings, and content distribution across different countries.

The project also focuses on cleaning and preprocessing the dataset, creating meaningful visualizations, and providing actionable insights based on the analysis.

3. Outcome

The analysis helps in identifying key trends in Netflix content growth and patterns in audience preferences.

It provides a clear understanding of content distribution, popular genres, and rating patterns.

The insights derived from this project can support better content strategy and decision-making.

3. Domain

Social Media & Public Sentiment

4. Dataset Information

Source: Kaggle Dataset Link : <https://www.kaggle.com/datasets/bhargavchirumamilla/netflix-movies-and-tv-shows-till-2025/data>

Year / Timeline: Data collected from Up to 2025

Dataset Size: The dataset contains more than 500 rows and more than 10 columns

Dataset Columns Description :

Title : Name of the movie or TV show Type : Category (Movie or TV Show) Genre : Type of content (Drama, Comedy, etc.) Release Year : Year in which the content was released Rating : Audience rating category Duration : Length of the content Country : Country where the content was produced Date Added : Date when the content was added to Netflix Director : Director of the content Cast : Actors involved in the content

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5. Type of Analysis

Descriptive Analysis : Descriptive analysis is used to summarize the dataset and understand the overall distribution of Netflix content. It helps in identifying the total number of movies and TV shows, distribution of genres, ratings, and content across different countries and years.

Diagnostic analysis : Diagnostic analysis is used to understand the reasons behind observed patterns in the dataset. For example, it helps to analyze why certain genres are more popular, why specific countries produce more content, and what factors influence content distribution.

Predictive analysis : Predictive analysis is used to identify future trends in Netflix content growth based on past data. By analyzing year-wise trends, it helps in estimating how content production may grow in the future, such as predicting trends for upcoming years like 2026.

Prescriptive analysis : Prescriptive analysis is used to provide recommendations based on the insights obtained from the data. It helps in suggesting what type of content Netflix should focus on, which genres are performing well, and how content strategy can be improved.

Stages for DA Project

Stage 1 – Problem Definition and Dataset Selection

Define the business problem and expected outcome

Choose dataset and explain its source, size, and features

Import libraries, load dataset

Dataset description (rows, columns, features)

Initial EDA (head, info, describe, shape, null checks, duplicate check)

```
import pandas as pd
```

```
from google.colab import files  
uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session.
Please rerun this cell to enable.

```
df = pd.read_csv("https://raw.githubusercontent.com/Mohanavalli-369/mydataset/refs/heads/main/netflix_  
movies_detailed_up_to_2025.csv.csv")
```

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df.head()

df.head()																	
...	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres	language	description	popularity	vote_count	vote_average	
0	10192	Movie	Shrek Forever After	Mike Mitchell	Mike Myers, Eddie Murphy, Cameron Diaz, Antoni...	United States of America	2010-05-16	2010	6.380	NaN	Comedy, Adventure, Fantasy, Animation, Family	en	A bored and domesticated Shrek pacts with deal...	203.893	7449	6.380	1
1	27205	Movie	Inception	Christopher Nolan	Leonardo DiCaprio, Joseph Gordon-Levitt, Ken W...	United Kingdom, United States of America	2010-07-15	2010	8.369	NaN	Action, Science Fiction, Adventure	en	Cobb, a skilled thief who commits corporate es...	156.242	37119	8.369	1
2	12444	Movie	Harry Potter and the Deathly Hallows: Part 1	David Yates	Daniel Radcliffe, Emma Watson, Rupert Grint, T...	United Kingdom, United States of America	2010-11-17	2010	7.744	NaN	Adventure, Fantasy	en	Harry, Ron and Hermione walk away from their l...	121.191	19327	7.744	2
3	38757	Movie	Tangled	Byron Howard, Nathan Greno	Mandy Moore, Zachary Levi, Donna Murphy,	United States of America	2010-11-24	2010	7.600	NaN	Animation, Family, Adventure	en	Feisty teenager Rapunzel, who has long and mag...	111.762	11638	7.600	2

df.shape

...(16000, 18)

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 16000 entries, 0 to 15999

Data columns (total 18 columns):

Column Non-Null Count Dtype

```
---
0 show_id      16000 non-null int64
1 type         16000 non-null object
2 title        16000 non-null object
3 director     15868 non-null object
4 cast         15796 non-null object
5 country      15534 non-null object
6 date_added   16000 non-null object
7 release_year 16000 non-null int64
8 rating       16000 non-null float64
9 duration     0 non-null float64
10 genres      15893 non-null object
11 language    16000 non-null object
12 description 15868 non-null object
13 popularity  16000 non-null float64
14 vote_count   16000 non-null int64
15 vote_average 16000 non-null float64
16 budget      16000 non-null int64
17 revenue     16000 non-null int64
```

dtypes: float64(4), int64(5), object(9)

memory usage: 2.2+ MB

df.describe()

df.describe()									
...	show_id	release_year	rating	duration	popularity	vote_count	vote_average	budget	revenue
count	1.600000e+04	16000.000000	16000.000000	0.0	16000.000000	16000.000000	16000.000000	1.600000e+04	1.600000e+04
mean	5.266582e+05	2017.500000	5.956368	NaN	20.384728	718.656125	5.956368	8.766792e+06	2.446308e+07
std	3.794427e+05	4.609916	1.754741	NaN	68.610033	2080.198316	1.754741	2.912450e+07	1.116977e+08
min	1.890000e+02	2010.000000	0.000000	NaN	3.860000	0.000000	0.000000	0.000000e+00	0.000000e+00
25%	2.257258e+05	2013.750000	5.600000	NaN	7.840750	53.000000	5.600000	0.000000e+00	0.000000e+00
50%	4.468170e+05	2017.500000	6.300000	NaN	10.913500	138.000000	6.300000	0.000000e+00	0.000000e+00
75%	7.739808e+05	2021.250000	6.923000	NaN	17.336500	422.000000	6.923000	2.200000e+06	1.654473e+06
max	1.440471e+06	2025.000000	10.000000	NaN	3876.006000	37119.000000	10.000000	4.600000e+08	2.799439e+09

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```
df.isnull().sum()
```

```
0
show_id0
type0
title0
director132
cast204
country466
date_added0
release_year0
rating0
duration16000
genres107
language0
description132
popularity0
vote_count0
vote_average0
budget0
revenue0
```

```
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

Stage 2 – Data Cleaning and Pre-processing

Handle missing values (impute or drop)

Handle duplicates

column name rename or dropping unwanted columns if required

Treat outliers if required

Check skewness and apply transformations

Convert data types if needed

Feature transformations (date parts, derived fields if required for analysis)

```
# Filling duration with Unknown because it is a categorical/text column
df['duration'] = df['duration'].fillna("Unknown")
```

```
# Filling categorical columns with meaningful values
df['country'] = df['country'].fillna("Unknown")
df['genres'] = df['genres'].fillna("Not Specified")
```

```
# Filling text columns to avoid data loss
df['cast'] = df['cast'].fillna("Not Available")
df['director'] = df['director'].fillna("Not Available")
```

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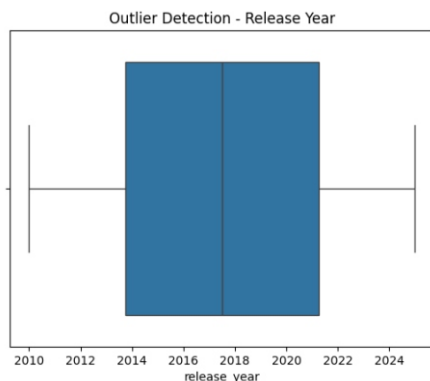
```
# Dropping description column as it is not required for analysis
df = df.drop(['description'], axis=1, errors='ignore')
```

```
# Checking duplicate records
df.duplicated().sum()
# No duplicate records found in this dataset
```

```
np.int64(0)
```

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Checking outliers using boxplot
sns.boxplot(x=df['release_year'])
plt.title("Outlier Detection - Release Year")
plt.show()
```



A boxplot was used to detect outliers in the release year column. The visualization shows that there are no significant outliers present in the dataset no outlier treatment was required

```
# Data Type Conversion
# Converting date added to datetime format for better time based analysis
df['date_added'] = pd.to_datetime(df['date_added'])
```

```
# Creating new column year added for analysis
df['year_added'] = df['date_added'].dt.year
```

```
# Filtering dataset to include recent data (2022 onwards)
df = df[df['release_year'] >= 2022]
```

```
df.to_csv("cleaned_netflix_data.csv")
```

```
from google.colab import files
files.download("cleaned_netflix_data.csv")
```

```
df.to_excel("cleaned_netflix_data.xlsx")
```

```
from google.colab import files
files.download("cleaned_netflix_data.xlsx")
```

NETFLIX ANALYTICS: EXPLORING CONTENT GROWTH AND PATTERNS

```
# Dropping description column as it is not required for analysis
df = df.drop(['description'], axis=1, errors='ignore')
```

In this stage the dataset was cleaned and preprocessed to improve data quality. Missing values were handled by filling categorical and text columns with appropriate labels such as "Unknown" and "Not Available" to avoid data loss.

Duplicate records were checked and no duplicates were found in the dataset. Unnecessary columns were removed to simplify the analysis.

Outliers were checked using a boxplot, and no significant outliers were found so no treatment was required.

The date column was converted into datetime format, and a new feature called yearadded was created for better analysis.

The duration column contained textbased values, so missing values were handled by filling them with Unknown instead of using numerical methods.

Finally the dataset was filtered to include recent data for further analysis.

Stage 3 – EDA and Visualizations

Univariate Analysis → distribution of single variables (countplot, histogram, boxplot)

Bivariate Analysis → relation between two variables (scatterplot, barplot, correlation heatmap)

Multivariate Analysis → relation among 3+ variables (pairplot, grouped analysis, pivot tables, advanced plots)

Interpretation MUST with every visualization

Focus on business story not just charts

Each chart must be in separate cells.

Rating Distribution

Insights :

Most content has ratings between 5 and 7

Very few content items have extremely high ratings

Some content has very low ratings (outliers)

Ratings are concentrated in the mid-range

Interpretation of above chart :

Explain the Chart.

This chart shows the distribution of average ratings using a histogram.

What is it saying?

It shows that most content has moderate ratings, with fewer items having very high or very low ratings

What features you used?

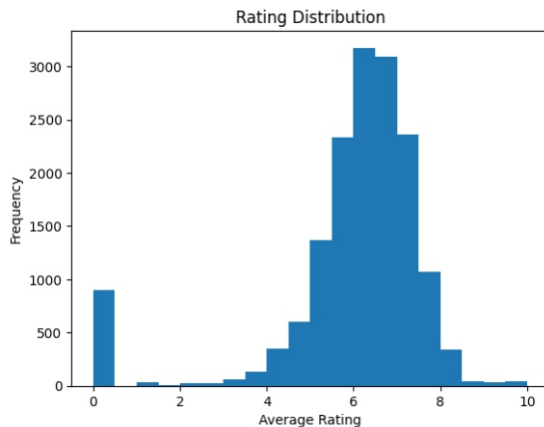
The feature used is vote average.

From this what you are showing us?

This indicates that most Netflix content performs at an average level rather than extremely high or low

NETFLIX ANALYTICS: EXPLORING CONTENT GROWTH AND PATTERNS

```
plt.hist(df['vote_average'], bins=20)
plt.title("Rating Distribution")
plt.xlabel("Average Rating")
plt.ylabel("Frequency")
plt.show()
```



Average Rating Over Years

Insights :

Average ratings are mostly stable across all years

Slight increase in ratings can be seen after 2016

No major fluctuations in ratings over time

Content quality remains consistent across years

Interpretation of above chart :

Explain the Chart.

This chart shows the average rating of content across different release years.

What is it saying?

It shows that ratings are mostly stable with slight improvements in recent years.

What features you used?

The features used are release_year and vote average

From this what you are showing us?

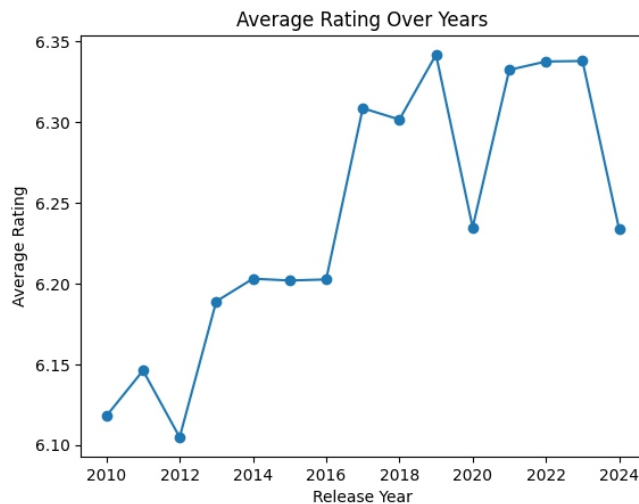
This indicates that content quality has remained consistent over time with minor improvements

```
# remove last year (example: 2025)
filtered_df = df[df['release_year'] < 2025]

year_rating = filtered_df.groupby('release_year')['vote_average'].mean()

plt.plot(year_rating.index, year_rating.values, marker='o')
plt.title("Average Rating Over Years")
plt.xlabel("Release Year")
plt.ylabel("Average Rating")
plt.show()
```


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Top Countries Producing Content

Insights :

The United States produces the highest amount of content

A few countries dominate Netflix content production

Content is not evenly distributed across all countries

Interpretation of above chart :

Explain the Chart.

This chart shows the top countries producing content available on netflix

What is it saying?

It shows that the United States contributes the highest number of titles compared to other countries

What features you used?

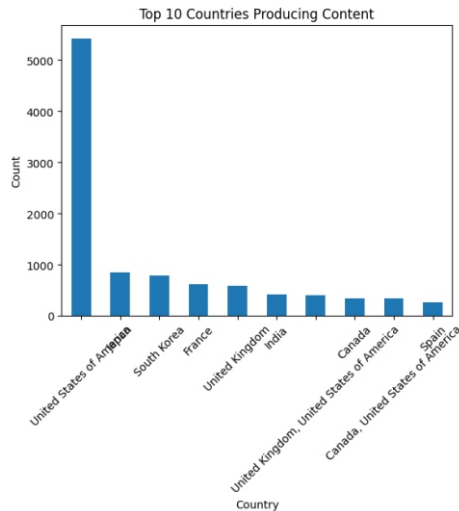
The feature used here is the country column

From this what you are showing us?

This indicates that Netflix content production is concentrated in a few countries especially the United States.

```
dff['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Countries Producing Content")
plt.xlabel("Country")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```

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Popularity vs Rating

Insights :

Higher ratings generally lead to higher popularity

Some content is popular even with moderate ratings

There is a positive relationship between rating and popularity

Interpretation of above chart :

Explain the Chart.

This chart shows the relationship between average rating and popularity of netflix content.

What is it saying?

It shows that content with higher ratings tends to have higher popularity.

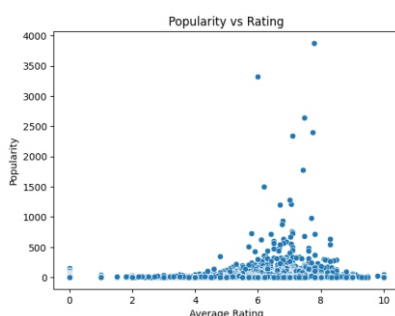
What features you used?

The features used are vote_average and popularity.

From this what you are showing us?

This indicates that audience ratings have an impact on content popularity

```
sns.scatterplot(x='vote_average', y='popularity', data=df)
plt.title("Popularity vs Rating")
plt.xlabel("Average Rating")
plt.ylabel("Popularity")
plt.show()
```



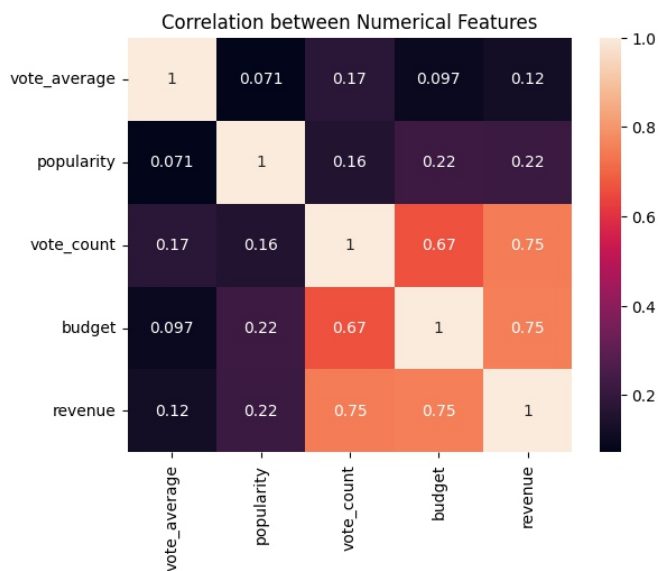
Correlation Heatmap

- Insights :**
- vote count and revenue have a strong positive correlation (0.75)
 - budget and revenue also show a strong positive relationship (0.75)
 - vote count and budget have a moderately strong relationship (0.67)
 - popularity has weak correlation with most variables
 - vote average has very weak correlation with all other features

Interpretation of above chart :

- Explain the Chart.
- This chart shows the correlation between numerical features using a heatmap.
- What is it saying?
- It shows that revenue is strongly related to budget and vote count while rating and popularity have weak relationships with other variables.
- What features/columns you used?
- The features used are vote average, popularity, vote count, budget and revenue
- From this what you are showing us?
- This indicates that financial and engagement metrics (budget, vote count) have a strong impact on revenue while ratings do not strongly influence revenue

```
sns.heatmap(df[['vote_average','popularity','vote_count','budget','revenue']].corr(), annot=True)
plt.title("Correlation between Numerical Features")
plt.show()
```



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Vote Count Distribution

Insights :

Most content has very low vote counts

There are many outliers with extremely high vote counts

The data is highly right-skewed

Only a few content items receive very high audience engagement

Interpretation of above chart :

Explain the Chart.

This chart shows the distribution of vote counts using a boxplot

What is it saying?

It shows that most content has low vote counts, while a few content items have extremely high vote counts

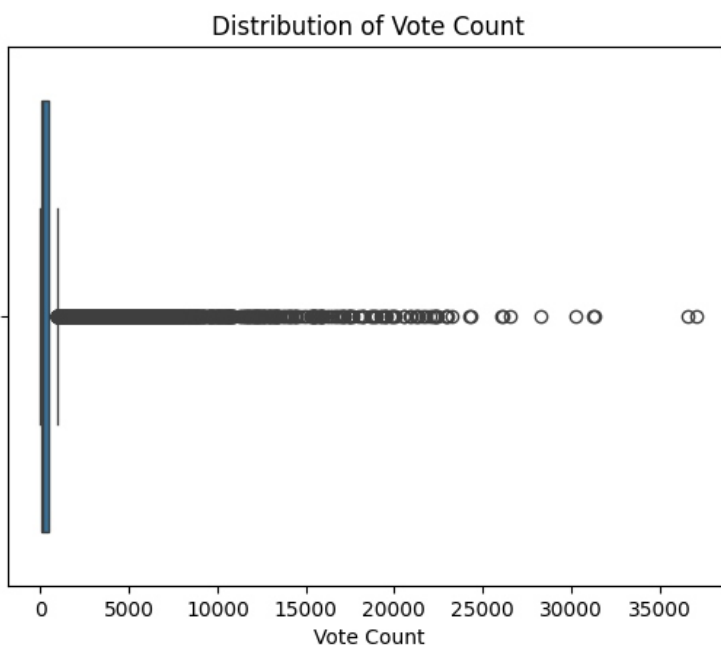
What features/columns you used?

The feature used is vote count.

From this what you are showing us?

This indicates that audience engagement is concentrated on a small number of popular content items, while most content receives limited engagement

```
sns.boxplot(x=df['vote_count'])  
plt.title("Distribution of Vote Count")  
plt.xlabel("Vote Count")  
plt.show()
```



Log Transformed Vote Count Distribution

Insights :

After log transformation, the distribution becomes more balanced

Most content falls within the mid-range of vote counts

Extreme high values are reduced and easier to visualize

The data shows a more normal-like distribution after transformation

Interpretation of above chart :

Explain the Chart.

This chart shows the distribution of vote counts after applying log transformation

What is it saying?

It shows a more balanced and spread-out distribution compared to the original highly skewed data

What features/columns you used?

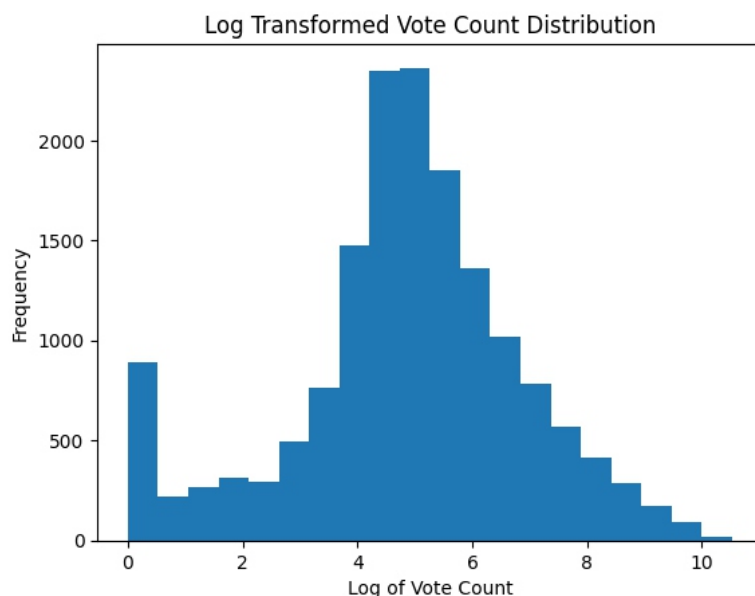
The feature used is vote_count, after applying log transformation

From this what you are showing us?

This indicates that log transformation helps in reducing skewness and makes the data easier to analyze and interpret.

```
import numpy as np

plt.hist(np.log1p(df['vote_count']), bins=20)
plt.title("Log Transformed Vote Count Distribution")
plt.xlabel("Log of Vote Count")
plt.ylabel("Frequency")
plt.show()
```



NETFLIX ANALYTICS: EXPLORING CONTENT GROWTH AND PATTERNS

Top 10 Highest Revenue Content

Insights :

A few content items generate extremely high revenue

Avengers: Endgame has the highest revenue among all

Top movies dominate the revenue compared to others

Revenue is concentrated among blockbuster movies

Interpretation of above chart :

Explain the Chart.

This chart shows the top 10 highest revenue-generating content using a bar chart.

What is it saying?

It shows that a small number of movies generate significantly higher revenue than others

What features/columns you used?

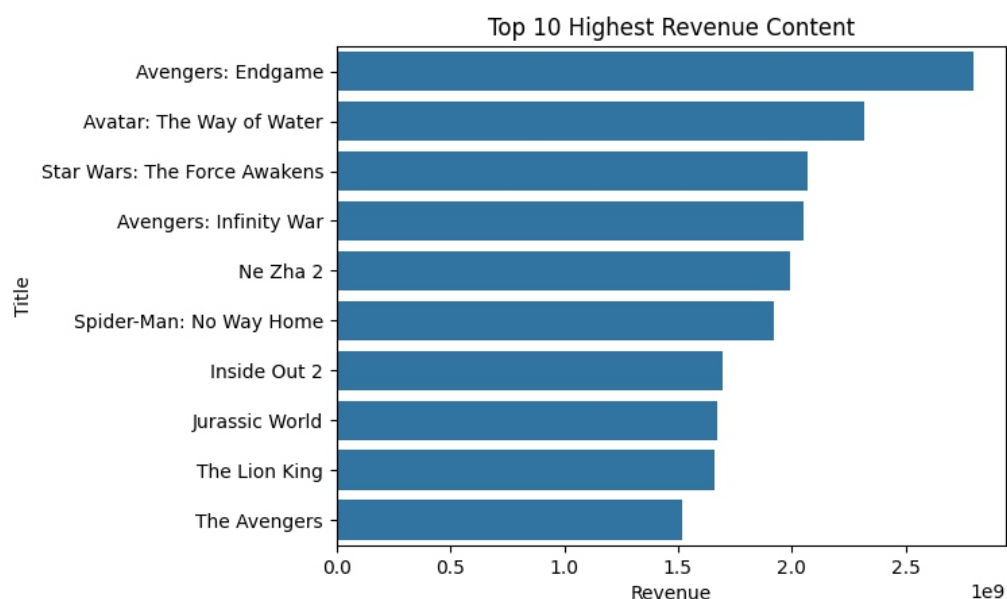
The features used are title and revenue

From this what you are showing us?

This indicates that revenue is highly concentrated among blockbuster content, while most other content generates comparatively lower revenue.

```
top_revenue = df.sort_values(by='revenue', ascending=False).head(10)

sns.barplot(x=top_revenue['revenue'], y=top_revenue['title'])
plt.title("Top 10 Highest Revenue Content")
plt.xlabel("Revenue")
plt.ylabel("Title")
plt.show()
```



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In this stage EDA was performed using different charts to understand the data clearly.

Univariate analysis was used to study individual variables like ratings, vote count, and popularity.

Bivariate analysis was used to understand relationships between variables such as rating vs popularity and revenue vs budget.

A heatmap was used for multivariate analysis to find relationships between multiple numerical features.

From the analysis, it was observed that most content has moderate ratings and low popularity. Only a few content items have very high popularity and revenue.

It was also found that budget and vote count have a strong relationship with revenue.

Log transformation was applied to vote_count to reduce skewness and make the data easier to understand.

Overall the charts helped to identify patterns, relationships, and important insights from the data.

Steps to Download your cleaned data for dashboard creation using Power BI

```
import pandas as pd
df=pd.read_csv("https://raw.githubusercontent.com/geethag-sys/Health-care-analysis3/refs/heads/main/insurance.csv")
df.to_csv("insurance.csv")
```

Stage 4 – Documentation, Insights and Presentation

Handle missing values (impute or drop)

Handle duplicates

column name rename or dropping unwanted columns if required

Treat outliers if required

Check skewness and apply transformations

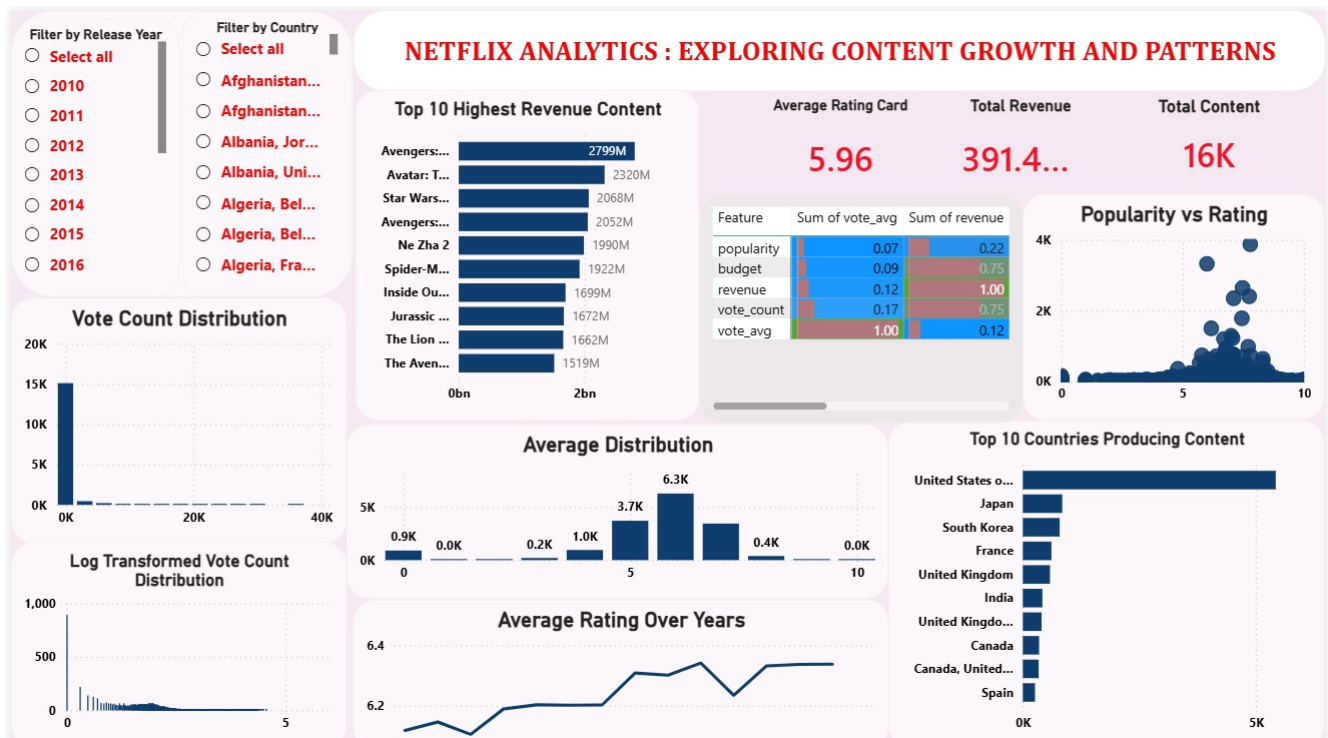
Convert data types if needed

Feature transformations (date parts, derived fields if required for analysis)

Power BI Dashboard Screenshot

NETFLIX ANALYTICS: EXPLORING CONTENT GROWTH AND PATTERNS

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Dashboard Explanation

This dashboard provides an overview of Netflix content trends, including ratings, popularity, revenue, and content distribution. It helps in understanding how content performs across different factors such as time, country, and audience engagement.

Overall Insights

- Most content has average ratings between 5 and 7
- Content quality is consistent over the years
- The United States dominates content production
- Budget and audience engagement strongly influence revenue
- Only a few titles receive high popularity and votes

Business Recommendations

- Focus on high-budget content to increase revenue
- Improve marketing for mid-rated content
- Expand production in different countries

Final Story with Dashboard

This dashboard shows that most Netflix content has average ratings and consistent performance over time. A few countries dominate content production, and only a small number of titles generate high revenue and popularity. Budget and audience engagement play a key role in determining success. Overall, the data highlights that content performance depends on multiple factors rather than ratings alone.

Conclusion

The analysis shows that most Netflix content has average ratings and consistent quality over time. A few countries dominate content production, and financial factors like budget and audience engagement play a key role in revenue generation. Overall, only a small number of titles achieve high popularity and success.